5. Spacetime is Quantum

There is a paradox at the heart of our understanding of the physical world. General relativity and quantum mechanics, the two jewels that the twentieth century has left us, have been prolific in gifts – for comprehending the world and for today's technology. From the first of these, cosmology has developed, as well as astrophysics, the study of gravitational waves and of black holes. The second has provided the foundation for atomic physics, nuclear physics, the physics of elementary particles and of condensed matter, and of much else besides.

And yet between the two theories there is something that grates. They cannot both be true, at least not in their present forms, because they appear to contradict each other. The gravitational field is described without taking quantum mechanics into account, without accounting for the fact that fields are quantum fields – and quantum mechanics is formulated without taking into account the fact that spacetime curves and is described by Einstein's equations.

A university student attending lectures on general relativity in the morning, and others on quantum mechanics in the afternoon, might be forgiven for concluding that his professors are fools, or that they haven't talked to each other for at least a century. In the morning, the world is a *curved* spacetime where everything is *continuous*; in the afternoon, the world is a *flat* one where *discrete* quanta of energy leap and interact.

The paradox resides in the fact that both theories work remarkably well.

With every experiment and every test, nature continues to say 'you are right' to general relativity, and continues to say 'you are right' to quantum mechanics as well, despite the seemingly opposite assumptions on which the two theories are founded. It is clear that something still eludes us.

In most situations we can neglect quantum mechanics or general relativity (or both). The Moon is too large to be sensitive to minute quantum granularity, so we can forget the quanta when describing its movements. On the other hand, an atom is too light to curve space to a significant degree, and when we describe it we can forget the curvature of space. But there are situations where both curvature of space and quantum granularity matter, and for these we do not yet have an established physical theory that works.